3-12

Ozone Intrusion Event Associated with the East Asian Summer Monsoon: A Case Study Based on Daily Ozonesonde Measurements and Comparison to Chemical Reanalyses

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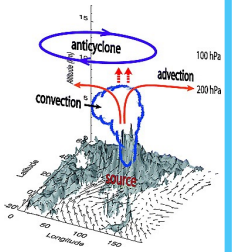
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Introduction

The Asian Summer Monsoon Anticyclone (ASMA) is a prominent atmospheric feature, playing a crucial role in regional and global climate systems. This ASMA significantly influences weather patterns and facilitates vertical air transport between the troposphere and stratosphere.

A key process associated with ASMA is the stratosphere-troposphere exchange (STE), where ozone-rich air from the stratosphere intrudes into the troposphere.



Park, M., et al, 2009

In this study, we used ozonesondes to observe the transport and variation of ozone during the Asian Summer Monsoon (ASM) period and compared the observational data with reanalysis data

O3sonde & reanalysis data

Table 1. Used data in this study.

Type	Data	Variable	Period	Resolution
Observation	ECC* Ozonesonde	O_3 MR*, O_3 PP*, TCO	2021. 08. 05 - 2021. 08. 31	~ 100 m
Atmospheric reanalysis	ERA5	GPH, PV, U, V		1.5°×1.5°, 6hr
Chemical reanalysis	MERRA-2	O_3 MR, O_3 PP, TCO, GPH, PV, U, V, T		0.5°×0.625° 3hr
	CAMS			0.75°×0.75° 3hr

*ECC: Electrochemical Concentration Cell
*MR: Mixing Ratio [ppbv]
*PP: Partial Pressure [mPa]



Observation Station

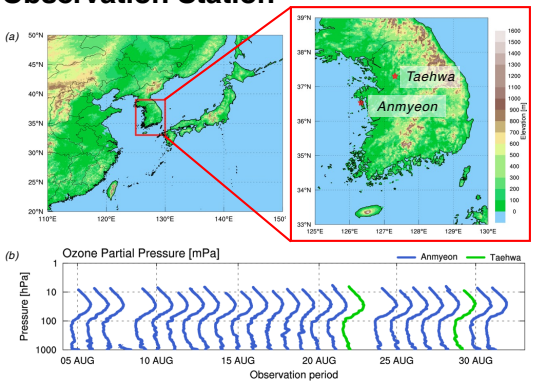


Figure 1. (a) Locations of Anmyeon (lat: 36.54°N, lon: 126.33°E) and Taehwa (lat: 37.31°N, lon: 127.31°E) ozonesonde observation sites in South Korea. Shading denotes elevation(unit: m). (b) Profiles of ozone partial pressure (unit: mPa) from ozonesonde measurements at Anmyeon (blue) and Taehwa (green) stations in South Korea during pre-ACCLIP period (August 2021).

Acknowledgments

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Observation results

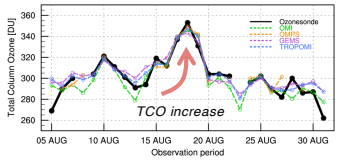


Figure 2. Time series of total column ozone (TCO; unit: DU) from ozonesonde measurements (black solid line with closed circles) and satellites (dashed line with open circles; OMI: orange, OMPS: orange, GEMS: purple, and TROPOMI: blue).

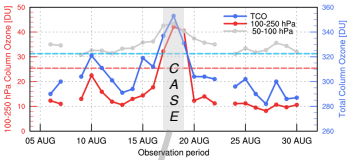


Figure 3. Time series of ozonesonde total (red solid line with dots), 100-250 hPa (blue solid line with dots), and 50-100 hPa (gray solid line with dots) column ozone (CO; unit: DU). Horizontal dashed lines are represented by the standard deviation (1 sigma) of ozonesonde total (orange, 25.4 DU) and 100-250 hPa (skyblue, 322.5 DU), respectively. Gray shading indicates stratosphere-troposphere exchange (STE) periods defined by the criterion of 100-250 hPa CO > 1 sigma.

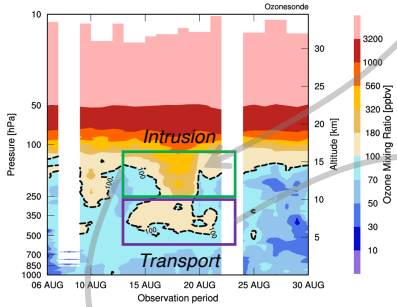


Figure 4. Time-height plot of ozone mixing ratio (shading, unit: ppbv) from ozonesonde measurements. Black dashed line indicates 100 ppbv.

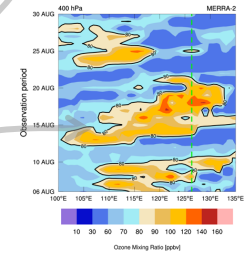


Figure 5. The Hovmöller diagram of the ozone mixing ratio (shading, unit: ppbv) from MERRA-2 at 400 hPa.

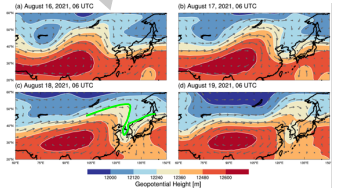


Figure 6. 200-hPa geopotential height (shading, unit: m) and wind (white vector, unit: m s^{-1}) from ERA5 reanalysis at August (a) 16, (b) 17, (c) 18, and (d) 19 2021, 06 UTC. The green line in (c) denotes anticyclonic wave breaking (AWB).

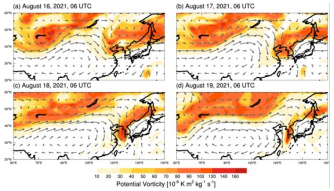


Figure 7. Same as in Figure 6, but for potential vorticity (unit: 10^{-6} K m^2 kg^{-1} s^{-1}).

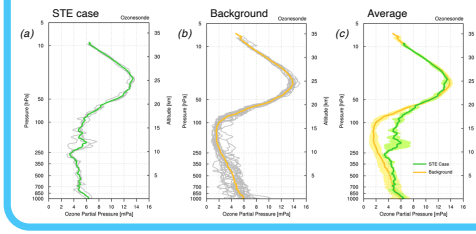


Figure 8. Profiles of ozone partial pressure(unit: mPa) from ozonesonde measurements for (a) case, (b) background, and (c) average. Thick green and orange lines indicate the mean profiles for the case and background, respectively. Thin gray lines show the individual profiles for (a) case and (b) background. Green and yellow shading in (c) indicate the standard deviation (1 sigma) of the case and background, respectively.

Comparison results

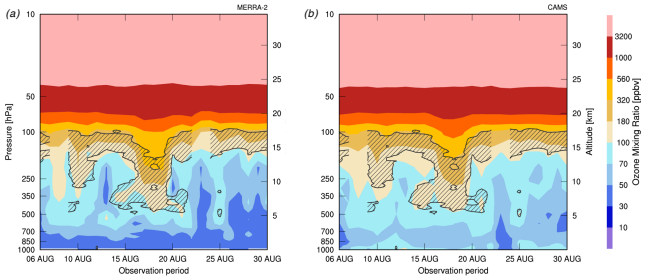


Figure 9. Same as in Figure. 4, but from chemical reanalyses of (a) MERRA-2 and (b) CAMS. The hatching zone indicates 100-320 ppbv of ozone mixing ratio from ozonesonde measurements.

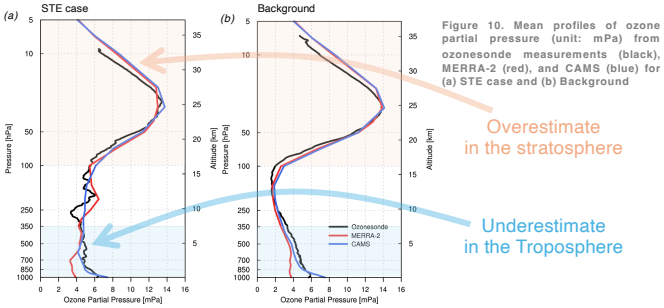


Figure 10. Mean profiles of ozone partial pressure (unit: mPa) from ozonesonde measurements (black), MERRA-2 (red), and CAMS (blue) for (a) STE case and (b) Background

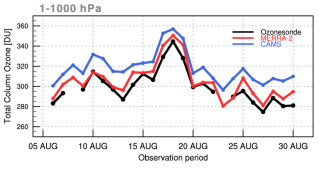


Figure 11. Time series of total column ozone from ozonesonde measurements (black), MERRA-2 (red), and CAMS (blue).

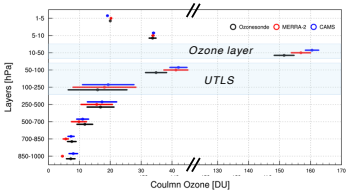


Figure 12. Layer column ozone distribution (unit: DU) from ozonesonde measurements (black), MERRA-2 (red), and CAMS (blue). The length of line denotes ±1σ on each layer.

Summary

- During the dry run of the ACCLIP season, stratosphere-troposphere exchange (STE) was observed due to the anticyclonic wave breaking.
- Both MERRA-2 and CAMS data indicate ozone intrusion and transport structures.
- The daily variability in TCO in MERRA-2 and CAMS closely matches that of the ozonesonde measurements.
- However, both MERRA-2 and CAMS significantly overestimate compared to ozonesonde in the UTLS and ozone layer regions.